



Chapter 4:

The Requirements (Business) Analyst



4.1 The Analyst Role

- Gathers, studies, documents, validates requirements
- Principal conduit between customer/user and developers
- Not a job title; software engineers, managers, domain experts, users can be analysts



4.2 The Analyst's Tasks, I

- Define business requirements
 - Understand why project is being done
 - Know project's vision and scope
- Identify stakeholders and users
 - Know who is affected by success or failure
 - Know who will interact directly with system



The Analyst's Tasks, 2

- Elicit requirements
 - proactively, persistently
 - variety of techniques—interviews, workshops, surveys, observations, work flow analysis,...
- Analyze requirements
 - Find derived, implied requirements
 - Find vagueness, conflicts



The Analyst's Tasks, 3

- Specify requirements
 - Clear enough for customer/user
 - Precise enough for developers
- Model requirements
 - Text usually isn't enough
 - Graphic models enable analysis
 - Prototypes increase understanding



The Analyst's Tasks, 4

- Validation
 - Lead the effort—schedule and chair the reviews
 - Ensure the excellence of the requirements
 - Ensure correct interpretation—review the code and test cases derived from requirements



The Analyst's Tasks, 5

- Facilitate prioritization—ensure successful negotiation between users and developers
- Manage requirements
 - Create the plan
 - Establish baselines
 - Control changes
 - Track status



4.3 Essential Skills, I

- **Listening**
 - Be attentive
 - Confirm understanding
 - Detect unspoken, implied needs
- **Interviewing and questioning**
 - Be able to talk with variety of people (even aggressive or intimidating)
 - Ask about normal and unusual uses and situations
 - Clarify uncertainties, disagreements, assumptions



Essential Skills, 2

- Analysis
 - Deal with abstraction and detail
 - Generalize and specialize
 - Think critically
 - Resolve conflicts from multiple sources
 - Separate wants from needs, solutions from requirements



Essential Skills, 3

- Facilitation
 - Be neutral
 - Relieve tension between customer/user and developers
 - Build trust on all sides
- Observation
 - See the subtle, in user words or actions



Essential Skills, 4

- Writing
 - Express complex ideas clearly
 - Have good command of customer/user language
 - Read efficiently and critically
- Organization
 - Structuring a large array of changing information into a coherent whole



Essential Skills, 5

- **Interpersonal relationships**
 - Getting competitors to work together
 - Mentor and educate
- **Creativity**
 - think of innovative capabilities
 - satisfy needs that user hasn't thought of yet



4.4 Essential Knowledge

- Modern techniques
 - Requirements engineering
 - Software development
- Project management
- Risk management
- Quality engineering
- Application domain knowledge



4.5 Making An Analyst, I

- Former User
 - Good understanding of business, work environment
 - Little familiarity with engineering or technical people
 - May not respect current users
 - May focus on user interface



Making An Analyst, 2

- **Former Developer**
 - Good technical skills
 - Often lacking in social, organizational skills
 - Little knowledge of business or application domain



Making An Analyst, 3

- Subject Matter (Domain) Expert
 - Excellent grasp of problem domain
 - Good at assessing reasonability and impacts on users
 - May prefer his own requirements over those of actual users
 - May create an unnecessarily ambitious system



END